Problem Statement

A Comparative Study of Urban Safety and Law Enforcement Resource Optimization in India (2020–2024)

Dataset Used

The analysis is based on the *Indian Crimes Dataset* sourced from Kaggle.

https://www.kaggle.com/datasets/sudhanvahg/indian-crimes-dataset

Data Preparation and Exploratory Data Analysis (EDA)

1) Dataset Overview

- The dataset consists of **40,160 records** and **14 attributes**.
- Missing values were observed in key fields:
    - **Weapons Used**: 5,790 missing entries
    - **Date Case Closed**: 20,098 missing entries

The high number of missing values in the *Date Case Closed* field indicates that nearly **50% of reported cases remain unresolved**, highlighting potential inefficiencies in crime resolution and reduced deterrence.

2) Handling missing values, inconsistencies observed:

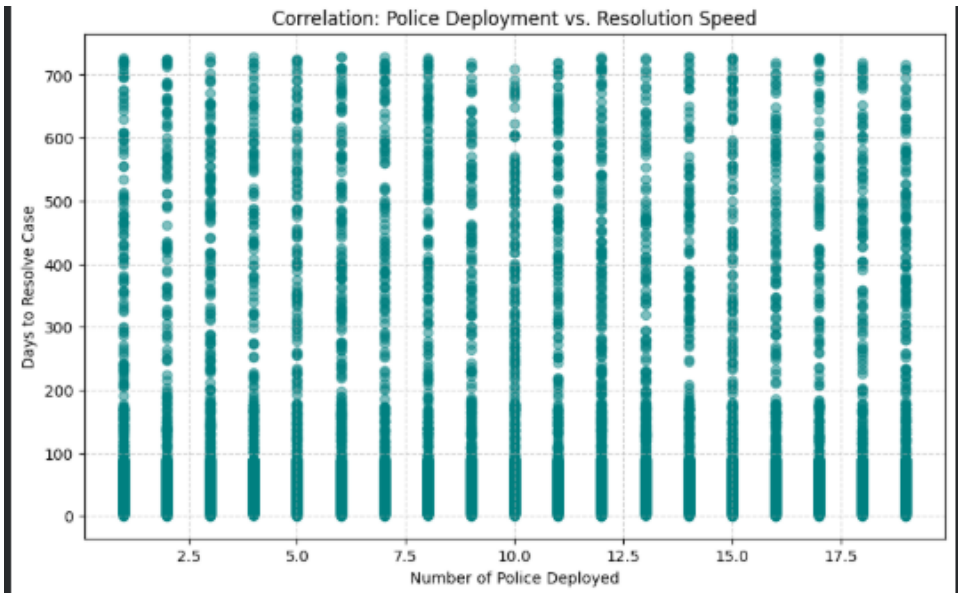
- Given the significant proportion of missing data, removal of such entries was avoided to **preserve data integrity and analytical depth**.
    - Missing values in *Weapons Used* were replaced with **"Unspecified"**, as the absence of information does not necessarily imply the absence of a weapon.
    - Since *Date Case Closed* contained nearly half the dataset as missing, these cases were retained and **analyzed as unresolved cases** to extract meaningful insights.
- Temporal inconsistencies were also identified across fields such as *Date Reported*, *Date of Occurrence*, and *Date Case Closed*.
- The **time component was excluded** due to its unreliable or placeholder nature.
- Dropped report number, crime code as they weren't much useful for the analysis. Similarly, dropped time of occurrence as it had inconsistencies in the time component.

3) Key Observations and Insights

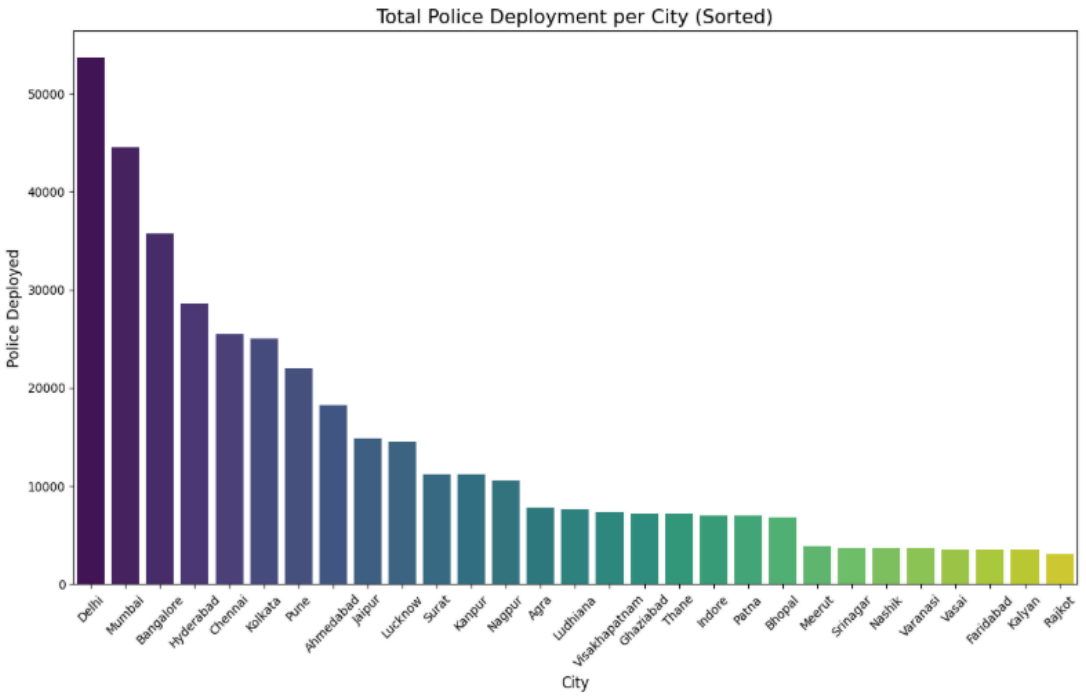
- The dataset includes approximately **500 distinct crime codes**, grouped into four broad categories:
    - Violent Crime
    - Fire Accident
    - Traffic Fatality
    - Other Crime
- The **'Other Crime' category dominates**, suggesting a diversification or evolution in crime patterns over time.
- The distribution of **weapons used** is relatively uniform, with each category (knife, explosives, blunt object, poison, firearm, and others) recording over **5,600 cases**, indicating no single dominant weapon type.
- **Urban concentration of crime** is evident, with metropolitan cities such as Mumbai and Delhi reporting significantly higher crime rates compared to suburban regions.
- **Gender-based analysis** reveals that **female victims constitute more than half of all reported cases**, indicating a heightened vulnerability.
- Further examination of cases labeled under "other" weapons showed **no meaningful differentiation in crime descriptions**, suggesting possible inconsistencies or lack of granularity in classification.

4) Analytical Insights

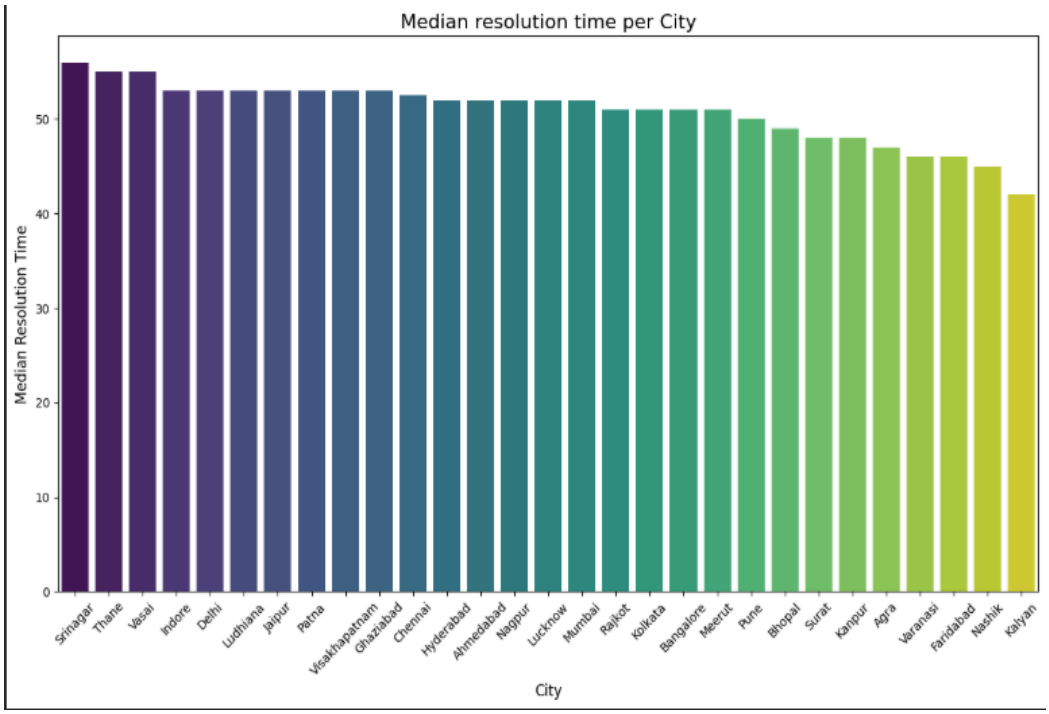
- There is **no significant correlation between police deployment and case resolution time**, suggesting inefficiencies in resource utilization.



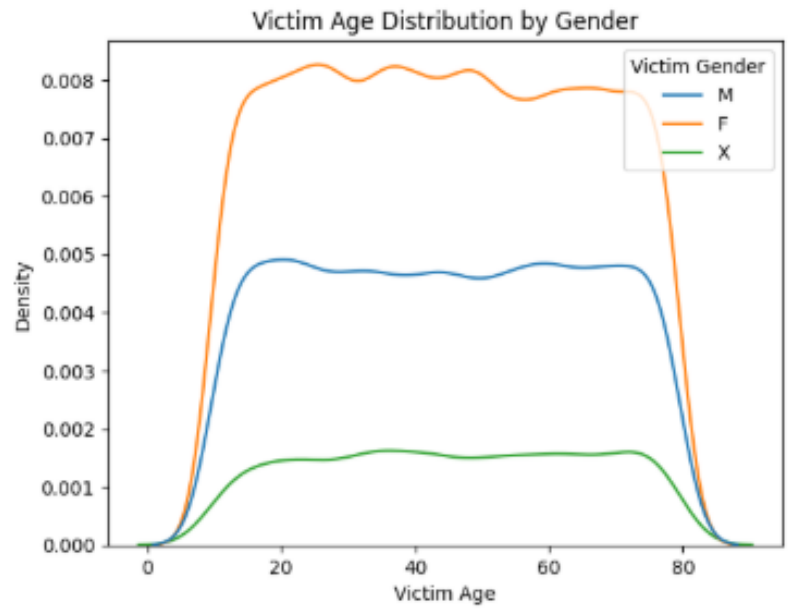
- A **positive correlation exists between city crime volume and police deployment**, indicating that higher crime rates lead to increased resource allocation.

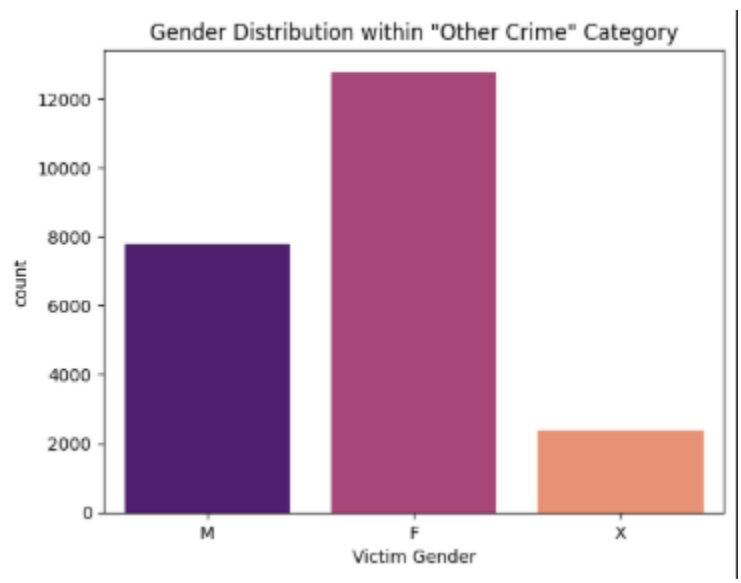


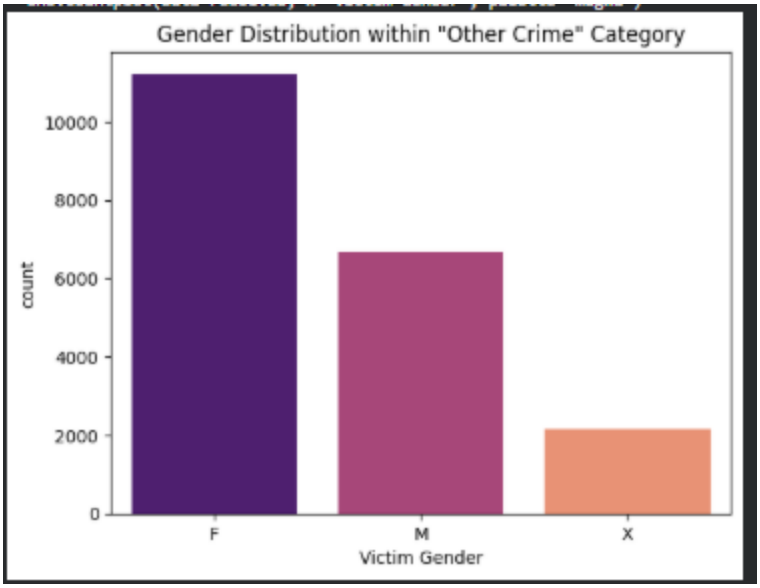
- Cities with high crime rates generally exhibit **moderate median resolution times**, likely due to proportional police deployment. However, cities such as Srinagar, Thane, and Vasai show **significantly higher resolution times**, indicating potential inefficiencies



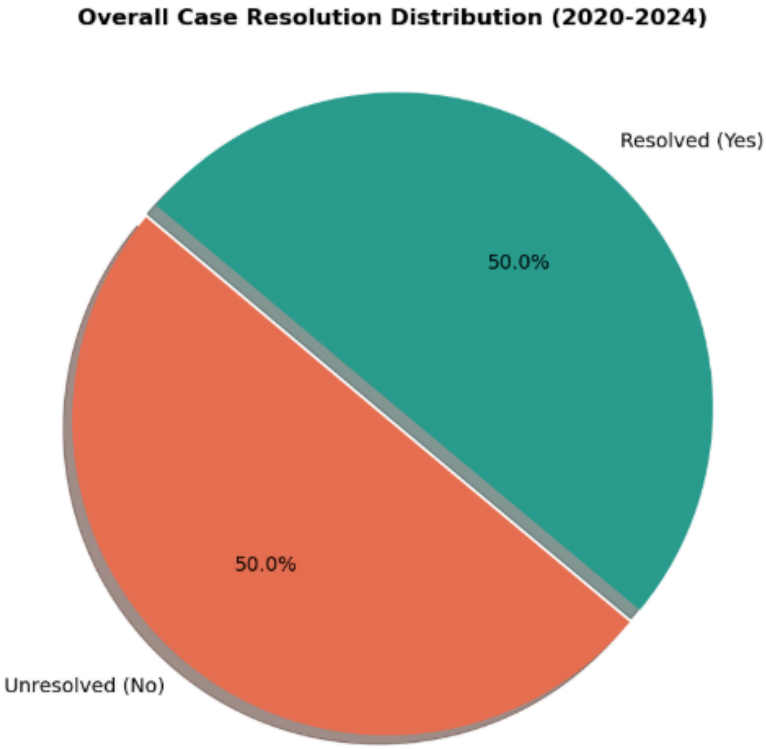
- Overall, women have always been a soft target towards the crimes, with more than half the reports being done for women victims.

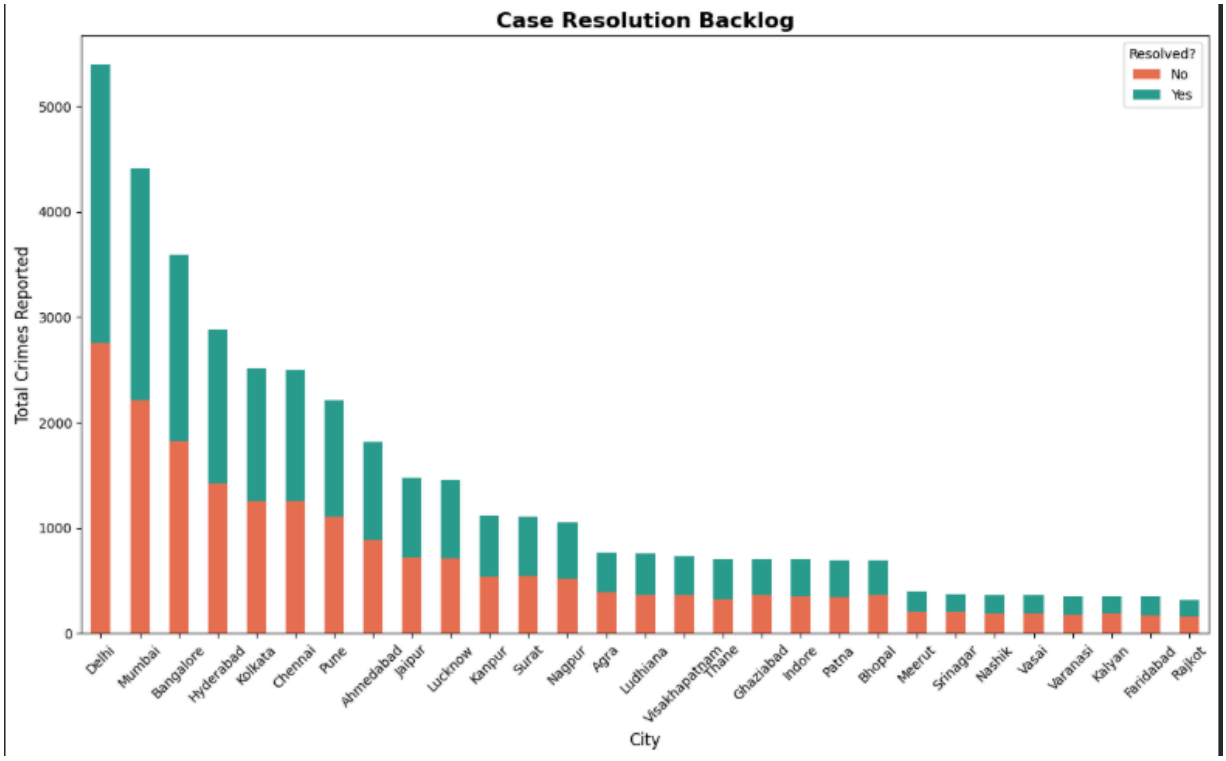

Victim Age Distribution by Gender
Victim Gender
M
F
X
Density
0.008
0.007
0.006
0.005
0.004
0.003
0.002
0.001
0.000
0
20
40
60
80
Victim Age


Gender Distribution within "Other Crime" Category
count
12000
10000
8000
6000
4000
2000
0
M
F
X
Victim Gender

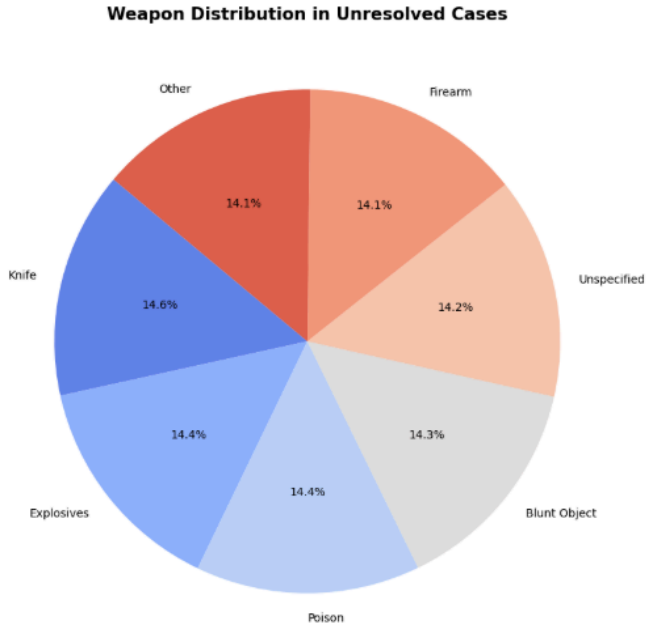


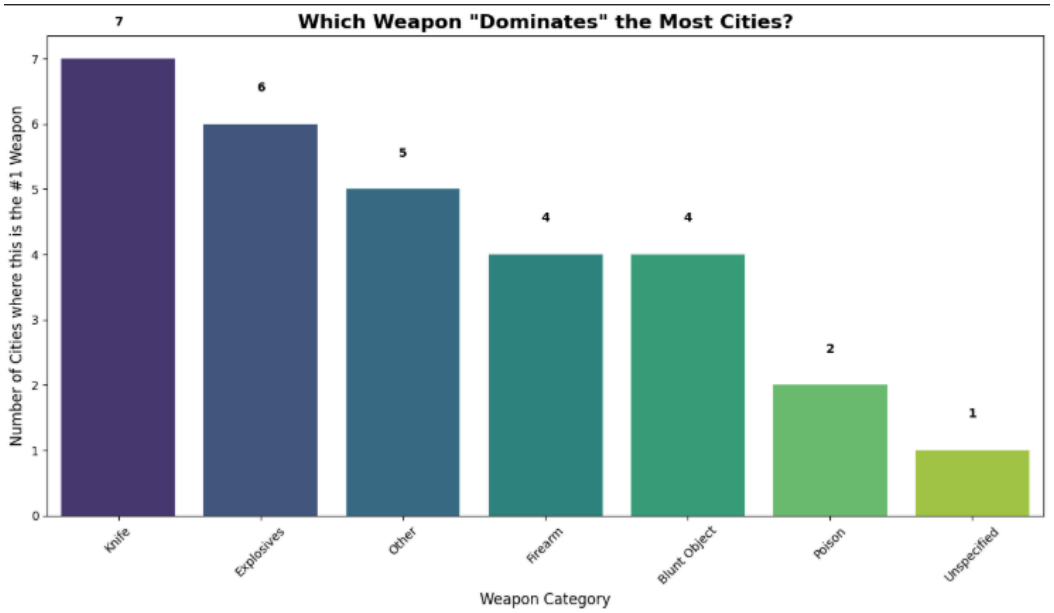
- The **median reporting lag across cities is approximately 1–2 days**, suggesting relatively prompt reporting behavior.
- At both national and city levels, **approximately half of all reported crimes remain unsolved**, reinforcing concerns regarding law enforcement effectiveness.





- While other weapons account for (¼)th of the weapons used, it has been observed that there is an equal contribution of other crimes in unresolved cases.





Further, one city has unspecified (missing) weapon as the major weapon being used for crimes. Also, unresolved cases, too, account for ½ the cases reported, and thus completely neglecting other weapons is not a safe choice.

Conclusion:

The study reveals a critical imbalance between **crime reporting and crime resolution in urban India**. While citizens demonstrate relatively prompt reporting behavior, the fact that nearly half of all cases remain unresolved highlights systemic inefficiencies in law enforcement processes.

Although police deployment increases with rising crime levels, it does not translate into faster case resolution, indicating that **resource allocation alone is insufficient without improvements in investigation efficiency, coordination, and data management**.

The dominance of the “Other Crime” category and the prevalence of “Unspecified” weapons further suggest **limitations in data classification and reporting practices**, which may hinder effective policy-making.

Additionally, the disproportionate impact of crime on women underscores the need for **targeted safety measures and gender-sensitive policing strategies**.

Overall, improving urban safety in India requires a shift from **reactive policing to data-driven, optimized resource allocation**, supported by better data quality, enhanced investigative frameworks, and systemic reforms in law enforcement